Linear Systems Theory

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Module 2
Lecture 5

Math Preliminaries: Linear Algebra 4

Fundamental Subspaces of Matrix

Matrix Subspaces

Based on linear maps, there are fundamental vector subspaces associated with any $m \times n$ matrix $\mathbf{A}$:

- Following are the fundamental subspaces:
    1. Column space or image of a matrix - $\mathcal{C}(\mathbf{A})$
    2. Null space or kernel of a matrix - $\mathcal{N}(\mathbf{A})$
    3. Row space of a matrix - $\mathcal{R}(\mathbf{A})$
    4. Left null space or kernel of transpose of a matrix - $\mathcal{N}(\mathbf{A}^T)$
- They are subspaces of either $\mathbb{R}^m$ or $\mathbb{R}^n$
- For $n \times n$ square matrices, all fundamental subspaces lie in the same vector space

Column space: $\mathcal{C}(\mathbf{A})$

Column space is the vector subspace spanned by columns of a matrix

- $\mathcal{C}(\mathbf{A})$ is a vector space with columns of $\mathbf{A}$ as its basis
- Follows from the image of a linear map
- Alternately, for an $m \times n$ matrix $\mathbf{A}$, column space comprises of all vectors $\mathbf{c} = \mathbf{A}\mathbf{x} \;\forall\; \mathbf{x} \;\in \mathbb{R}^n$

$$\mathcal{C}(\mathbf{A}) = \text{span}\{\mathbf{A}_{c1}, \mathbf{A}_{c2}, \ldots, \mathbf{A}_{cn}\}$$
$$\mathcal{C}(\mathbf{A}) = \{\mathbf{c} \mid \mathbf{c} = \mathbf{A}\mathbf{x} \;\forall\; \mathbf{x} \in \mathbb{R}^n\}$$

- $\mathcal{C}(\mathbf{A})$ is a subspace of $\mathbb{R}^m$ because $\mathbf{c} \in \mathbb{R}^m$
- $\dim \mathcal{C}(\mathbf{A})$ is atmost $\min(m, n)$

Null space: $\mathcal{N}(A)$

Null space is the vector subspace generated by the set of vectors $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{0}$

$$\mathcal{N}(A) = \{\mathbf{x} \in \mathbb{R}^n | A\mathbf{x} = \mathbf{0}\}$$

- Follows from the kernel of a linear map
- As $\mathbf{x} \in \mathbb{R}^n$, the null space is a subspace of $\mathbb{R}^n$.
- Alternately, $\mathcal{N}(A)$ contains all the vectors that lose their identity when mapped from $\mathbb{R}^n$ to $\mathbb{R}^m$ via A.

Row space: $\mathcal{R}(\mathbf{A})$

Row space is the vector subspace spanned by rows of a matrix

- $\mathcal{R}(\mathbf{A})$ contains all possible linear combinations of rows of $\mathbf{A}$
- Alternately, for an $m \times n$ matrix $\mathbf{A}$, row space comprises of all vectors $\mathbf{r} = \mathbf{A}^T\mathbf{y} \; \forall \; \mathbf{y} \in \mathbb{R}^m$

$$\mathcal{R}(\mathbf{A}) = \text{span}\{\mathbf{A}_{r1}^T, \mathbf{A}_{r2}^T, \ldots, \mathbf{A}_{rm}^T\}$$
$$\mathcal{R}(\mathbf{A}) = \{\mathbf{r} \mid \mathbf{r} = \mathbf{A}^T\mathbf{y} \; \forall \; \mathbf{y} \in \mathbb{R}^m\}$$

- $\mathcal{R}(\mathbf{A})$ is a subspace of $\mathbb{R}^n$ because $\mathbf{r} \in \mathbb{R}^n$
- Can also be considered as the image or range of $\mathbf{A}^T$

Left null space is the vector subspace generated by the set of vectors $\mathbf{y}$ such that $A^T\mathbf{y} = \mathbf{0}$

$$\mathcal{N}(A^T) = \{\mathbf{y} \in \mathbb{R}^m | A^T\mathbf{y} = \mathbf{0}\}$$

- As $\mathbf{y} \in \mathbb{R}^m$, the null space of A^T is a subspace of $\mathbb{R}^m$.
- Alternately, $\mathcal{N}(A^T)$ contains all the vectors that lose their identity when mapped from $\mathbb{R}^m$ to $\mathbb{R}^n$ via A^T.

Rank and Nullity

Rank

Rank of a linear map $f : U \to V$ with corresponding $m \times n$ matrix $\mathbf{A}$ is:

- the dimension of the image of f i.e., $\text{rank}\,(f) = \dim \mathbf{Im}(f)$
- the dimension of the column space of $\mathbf{A}$ (column rank) i.e., $\text{rank}\,(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A})$
- the dimension of the row space of $\mathbf{A}$ (row rank) i.e., $\text{rank}\,(\mathbf{A}) = \dim \mathcal{R}(\mathbf{A})$

$$r = \text{rank}\,(f) = \text{rank}\,(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A}) = \dim \mathcal{R}(\mathbf{A}) \leq \min(m, n)$$

- Rank of $\mathbf{A}$ gives the number of linearly independent columns or number of linearly independent rows of $\mathbf{A}$

Properties of Matrix Rank

- Matrix is of full rank if: $\mathrm{rank}\,(A) = \min(m, n)$
- Matrix is rank deficient if: $\mathrm{rank}\,(A) < \min(m, n)$
- $\mathrm{rank}\,(A) = \mathrm{rank}\,(A^T)$
- An $n \times n$ square matrix is invertible iff its rank is n
- $\mathrm{rank}\,(AB) \leq \min(\mathrm{rank}\,(A)\,, \mathrm{rank}\,(B))$

Nullity of Matrix

Nullity of a linear map $f : U \to V$ with corresponding $m \times n$ matrix $\mathbf{A}$ is:

- the dimension of kernel of f
- the dimension of the null space of $\mathbf{A}$

$$\textbf{nullity}\ (f) = \textbf{nullity}\ (\mathbf{A}) = \dim \ker(f) = \dim \mathcal{N}(\mathbf{A})$$

- Nullity is 0 if the null space contains only zero vector
- Rank and nullity of a matrix can be determined from its row echelon form

Row Echelon Form of a Matrix

A matrix is in row echelon form (ref) when it satisfies the following conditions:

1. the first non-zero element in each row, called the leading entry, is 1
2. each leading entry is in a column to the right of the leading entry in the previous row
3. rows with all zero elements, if any, are below rows having a non-zero element

- Eg: $A_{3\times 4} = \begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$; $B_{3\times 3} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \end{bmatrix}$
- Rank of a matrix is equal to the number of non-zero rows in its row echelon form
- Nullity is number of columns minus the rank of the matrix

Theorem 1: Rank-Nullity Theorem

For a linear map $f : U \to V$ with corresponding $m \times n$ matrix $\mathbf{A}$:

$$\text{rank}\,(f) + \textbf{nullity}\,(f) = \dim U$$
$$\text{rank}\,(\mathbf{A}) + \textbf{nullity}(\mathbf{A}) = n$$

Proof. $\square$

- For a linear map $f : U \to V$ with corresponding $m \times n$ matrix $\mathbf{A}$:

$$\text{rank}\left(\mathbf{A}^T\right) + \textbf{nullity}(\mathbf{A}^T) = m$$

Table 1: Summary of Fundamental Subspaces

Name	Subspace of	Notation	Dimension
Column space	$\mathbb{R}^m$	$\mathcal{C}(\mathbf{A})$	r
Null space	$\mathbb{R}^n$	$\mathcal{N}(\mathbf{A})$	n-r
Row space	$\mathbb{R}^n$	$\mathcal{R}(\mathbf{A})$	r
Left null space	$\mathbb{R}^m$	$\mathcal{N}(\mathbf{A}^T)$	m-r

Fundamental Theorem of Linear Algebra

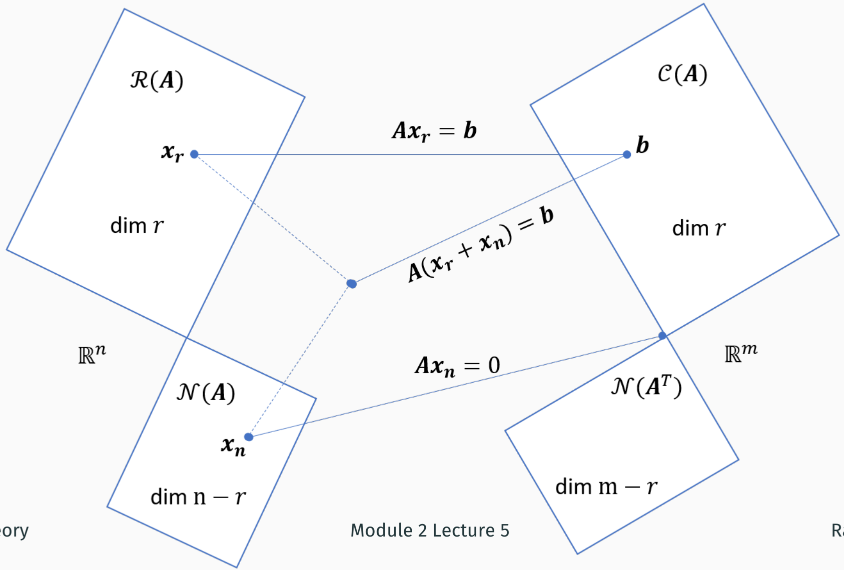
Theorem 2: Fundamental Theorem of Linear Algebra

1. The null space is orthogonal to the row space: In $\mathbb{R}^n, \mathcal{N}(\mathsf{A}) = \mathcal{R}(\mathsf{A})^\perp$
2. The left null space is orthogonal to the column space: In $\mathbb{R}^m, \mathcal{N}(\mathsf{A}^T) = \mathcal{C}(\mathsf{A})^\perp$

Proof.

$\square$

Visualisation of the Fundamental Subsapces



Overview

Summary: Mod 2 Lecture 5

- Fundamental subspaces of a matrix
- Rank and Nullity of a matrix
- Rank-Nullity Theorem
- Fundamental theorem of Linear Algebra

Contents: Module 3

- Change of Basis
- Types of Matrices
- Eigen Values and Eigen Vectors
- Invariant transformations /subspaces
- Similarly transformations